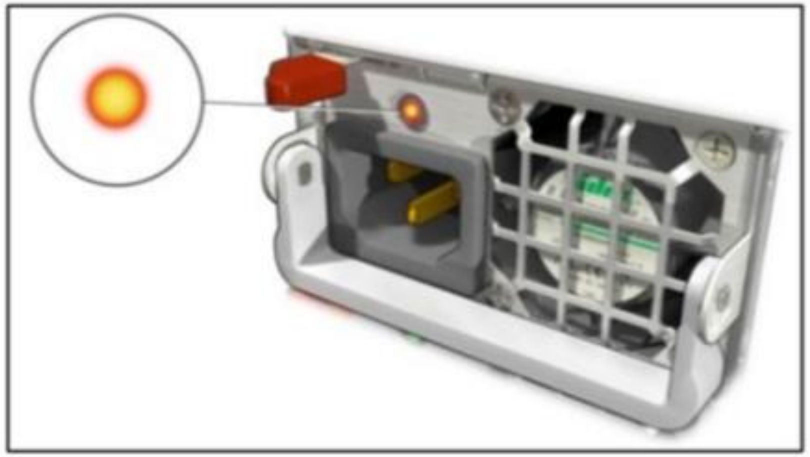
Power supply and fan troubleshooting

PSU and fan issues

Power supply diagnostics overview

To identify power supply issues, carry out the following actions:

- Physically look at the PSU indicator LEDs.
- Log in to Prism and go to **Hardware** → **Diagram** to check each server's power supply status.
- The `#ncc hardware_info show_hardware_info` command also shows the status of the PSU. (Click HERE to see the example output.)
- Go to the IMM/XCC interface.
- Use the host or CVM to run `ipmitool` commands. (Click HERE to see an example output.)



Physically check PSU LEDs

Use Prism to check the PSU

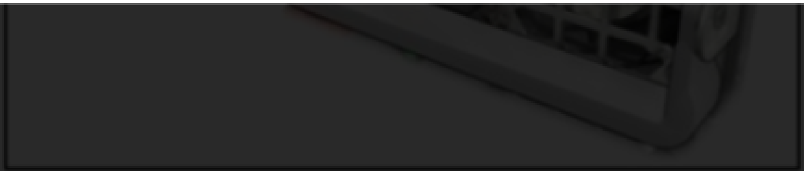
Power supply diagnostics overview

Output of #ncc hardware_info show_hardware_info

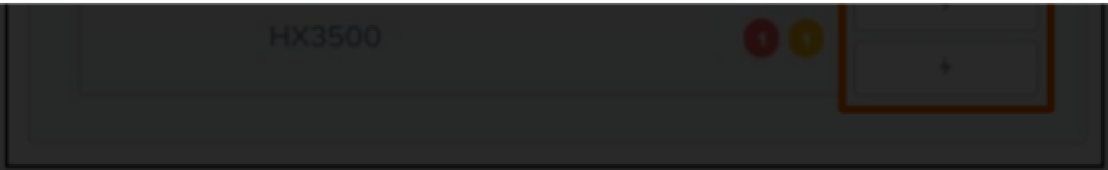
```
System Power Supply

    +-----------------------------------------------------------------------------------------+
    | Location                        |   PS                                                  |
    | Manufacturer                    |   ARTE                                                |
    | Max power capacity              |   0.0 W                                               |
    | Product part number             |   SP57A03100                                          |
    | Serial number                   |   P4LD83Y019F                                         |
    | Status                          |   Present, OK                                         |
    +-----------------------------------------------------------------------------------------+
    | Location                        |   PS                                                  |
    | Manufacturer                    |   ARTE                                                |
    | Max power capacity              |   0.0 W                                               |
    | Product part number             |   SP57A03100                                          |
    | Serial number                   |   P4LD83Y018J                                         |
    | Status                          |   Present, OK                                         |
    +-----------------------------------------------------------------------------------------+
```

ctions:



Physically check PSU LEDs



Use Prism to check the PSU

Power supply diagnostics overview

To identify power supply issues, carry out the following actions:

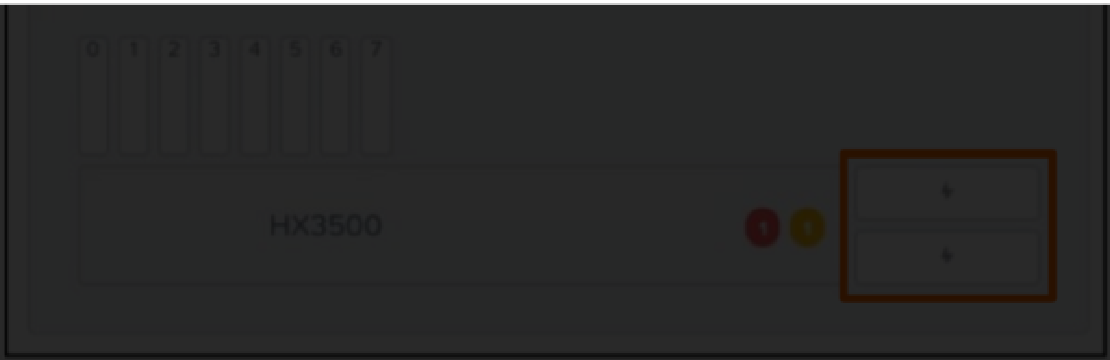
- Physically look at the PSU indicator LEDs.

re → **Diagram** to check each server's power supply

Output of #`ipmitool` command

```
ipmitool -H HOSTIP -U ADMIN -P ADMIN -v sel list
1 | 03/13/2014 | 02:47:10 | Power Supply #0xc8 | Failure detected | Asserted
2 | 03/13/2014 | 03:20:21 | Power Supply #0xc8 | Failure detected | Deasserted
3 | 03/13/2014 | 04:22:22 | Power Supply #0xc8 | Failure detected | Asserted
4 | 03/13/2014 | 05:33:57 | Power Supply #0xc8 | Failure detected | Deasserted
5 | 03/13/2014 | 23:19:20 | Power Supply #0xc8 | Failure detected | Asserted
6 | 03/14/2014 | 02:53:14 | Power Supply #0xc8 | Failure detected | Deasserted
7 | 03/18/2014 | 23:23:35 | Power Supply #0xc8 | Failure detected | Asserted
```

Physically check PSU LEDs



Use Prism to check the PSU

Fan diagnostics overview

- The `#ncc hardware_info show_hardware_info` command shows the status of the system fan. (Click HERE to see the example output.)
- The system fan speed should be set to **Standard Speed** in the system settings.
- Fan speed and status can be checked using `ipmitool`:
  - ESXi `#/ipmitool sensor list | grep -i FAN`
  - AHV `[root@NTNX-Block-1-A ~]# ipmitool sensor list | grep -i FAN`
  - Hyper-V Powershell `C:\> ipmiutil.exe sensor list | grep -i FAN`

```
[root@HX3500 ~]# ipmitool sensor list | grep -i FAN
Fan 1A Tach      | 4602.000   | RPM        | ok    | na    | 590.000    | na    | na    | na    | na
Fan 1B Tach      | 4410.000   | RPM        | ok    | na    | 630.000    | na    | na    | na    | na
Fan 2A Tach      | 4602.000   | RPM        | ok    | na    | 590.000    | na    | na    | na    | na
Fan 2B Tach      | 4473.000   | RPM        | ok    | na    | 630.000    | na    | na    | na    | na
Fan 3A Tach      | 4602.000   | RPM        | ok    | na    | 590.000    | na    | na    | na    | na
Fan 3B Tach      | 4473.000   | RPM        | ok    | na    | 630.000    | na    | na    | na    | na
Fan 4A Tach      | 4602.000   | RPM        | ok    | na    | 590.000    | na    | na    | na    | na
Fan 4B Tach      | 4410.000   | RPM        | ok    | na    | 630.000    | na    | na    | na    | na
Fan 5A Tach      | 4602.000   | RPM        | ok    | na    | 590.000    | na    | na    | na    | na
Fan 5B Tach      | 4410.000   | RPM        | ok    | na    | 630.000    | na    | na    | na    | na
Fan 6A Tach      | 4602.000   | RPM        | ok    | na    | 590.000    | na    | na    | na    | na
Fan 6B Tach      | 4410.000   | RPM        | ok    | na    | 630.000    | na    | na    | na    | na
Fan Riser1A Tach | 6256.000   | RPM        | ok    | na    | 3332.000   | na    | na    | na    | na
Fan Riser1B Tach | 6324.000   | RPM        | ok    | na    | 3332.000   | na    | na    | na    | na
Fan Riser2A Tach | na         |            | na    | na    | na         | na    | na    | na    | na
Fan Riser2B Tach | na         |            | na    | na    | na         | na    | na    | na    | na
```

Note: Users can also check system fan status via IMM/XCC.

Output of #ncc hardware_info show_hardware_info

Scroll down for more information.

```
FAN

   +----------------------------------------------------------------------------------+
   | Location                      |   Fan 1a tach                                    |
   | Rpm                           |   8988                                           |
   | Status                        |   OK                                             |
   +----------------------------------------------------------------------------------+
   | Location                      |   Fan 1b tach                                    |
   | Rpm                           |   10057                                          |
   | Status                        |   OK                                             |
   +----------------------------------------------------------------------------------+
   | Location                      |   Fan 2a tach                                    |
   | Rpm                           |   8988                                           |
   | Status                        |   OK                                             |
   +----------------------------------------------------------------------------------+
   | Location                      |   Fan 2b tach                                    |
   | Rpm                           |   10057                                          |
   | Status                        |   OK                                             |
   +----------------------------------------------------------------------------------+
   | Location                      |   Fan 3a tach                                    |
   | Rpm                           |   8988                                           |
   | Status                        |   OK                                             |
   +----------------------------------------------------------------------------------+
   | Location                      |   Fan 3b tach                                    |
   | Rpm                           |   10057                                          |
   | Status                        |   OK                                             |
```

Output of #ncc hardware_info show_hardware_info

Scroll down for more information.

```
| Location                    |    Fan 4a tach                                          |
| Rpm                         |    8988                                                 |
| Status                      |    OK                                                   |
+---------------------------------------------------------------------------------------+
| Location                    |    Fan 4b tach                                          |
| Rpm                         |    10057                                                |
| Status                      |    OK                                                   |
+---------------------------------------------------------------------------------------+
| Location                    |    Fan 5a tach                                          |
| Rpm                         |    8988                                                 |
| Status                      |    OK                                                   |
+---------------------------------------------------------------------------------------+
| Location                    |    Fan 5b tach                                          |
| Rpm                         |    10057                                                |
| Status                      |    OK                                                   |
+---------------------------------------------------------------------------------------+
| Location                    |    Fan 6a tach                                          |
| Rpm                         |    9660                                                 |
| Status                      |    OK                                                   |
+---------------------------------------------------------------------------------------+
| Location                    |    Fan 6b tach                                          |
| Rpm                         |    10057                                                |
| Status                      |    OK                                                   |
+---------------------------------------------------------------------------------------+
| Location                    |    Fan 7a tach                                          |
| Rpm                         |    9660                                                 |
```

Summary

This course enabled you to:

- Describe the procedure used to collect service logs with the Nutanix Prism Web console
- Describe how to manually perform a health check and show hardware information using commands
- Describe the general troubleshooting procedures for ThinkAgile HX Series hardware components